

Reviewer Pack — Minimal Number of Quasi-Crystals in Boolean Functions and Co...

Entry 4cf209a9cba5 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Number of Quasi-Crystals in Boolean Functions and Communication Complexity Rank Inequality

Entry ID: 4cf209a9cba5 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-05 07:50:33 UTC

1. Conjecture

Field A (mathematical branch): Quasi-Crystal Theory **Field B** (complexity object): Communication Complexity (Matrix Rank)

Statement:

For any given Boolean function f with n variables, the minimal number of quasi-crystals required to construct a representation of its associated matrix A is linearly correlated with its communication complexity rank $r(A)$, such that $|Q(f)| = \Theta(r(A))$

Rationale (proposer's reasoning):

Quasi-crystal theory provides a way to encode structures and patterns in complex systems, and its application to Boolean functions could provide new insights into the structure of communication complexity. This bridge might expose hidden symmetries or patterns in communication complexity that are not apparent through traditional methods.

Taxonomy category: Quasi-Crystal_Theoretic_Analysis (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 407ba7517405bcaa

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for at least 24 out of 30 seeds, $|Q(f)|/r(A)$ falls within $\pm 50\%$ of the expected $\Theta(r(A))$ ratio with a standard deviation ≤ 0.2 .

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_boolean_function(n):
        return [random.choice([0, 1]) for _ in range(2**n)]

    def matrix_from_function(f, n):
        A = [[f[i * (1 << n) + j] for i in range(1 << (n-1))] for j in range(1 << (n-1))]
        return A

    def communication_complexity_rank(A):
        m, n = len(A), len(A[0])
        rank = 0
        for i in range(m):
            if any(A[i][j] == 1 for j in range(n)):
                rank += 1
        return rank

    def quasi_crystals_required(A):
        m, n = len(A), len(A[0])
        count = 0
        for i in range(m):
            for j in range(n):
                if A[i][j] == 1:
                    count += 1
        return count

    n_values = [5, 10, 15, 20, 30, 40]
    results = []

    for n in n_values:
        f = generate_boolean_function(n)
        A = matrix_from_function(f, n)
        rank = communication_complexity_rank(A)
```

```

Q = quasi_crystals_required(A)

if rank == 0 or Q == 0:
    continue

results.append({
    "n": n,
    "Q": Q,
    "rank": rank
})

if not results:
    return {
        "metric_name": "Q/f vs rank",
        "metric_value": None,
        "instances_tested": 0,
        "n_max": 0,
        "conjecture_holds": False,
        "counterexample": "No valid instances found"
    }

Q_values = [r["Q"] for r in results]
rank_values = [r["rank"] for r in results]
mean_Q_over_rank = sum(Q / rank for Q, rank in zip(Q_values, rank_values)) / len(results)
std_dev = math.sqrt(sum((Q / rank - mean_Q_over_rank) ** 2 for Q, rank in zip(Q_values, rank_values)) / len(results))

return {
    "metric_name": "Q/f vs rank",
    "metric_value": mean_Q_over_rank,
    "instances_tested": len(results),
    "n_max": max(r["n"] for r in results),
    "conjecture_holds": abs(mean_Q_over_rank - 1) <= 0.5 and std_dev <= 0.2,
    "counterexample": ""
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [random.randint(1, 1000) for _ in range(30)]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    if all(r["conjecture_holds"] for r in results):
        mean_value = sum(r["metric_value"] for r in results) / len(results)
        support_fraction = 1.0
    else:
        mean_value = None
        support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["instances_tested"] >= 30 for r in results):
        if support_fraction >= 0.8:
            print(f"RESULT: SUPPORTED mean={mean_value} std=Unknown support_fraction={support_fraction}")
        else:

```

```

        first_failing_seed = next(i for i, r in enumerate(results) if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"Not enough instances supported\" first_fail={first_failing_seed}")
    else:
        n_tested = sum(r["instances_tested"] for r in results)
        print(f"RESULT: INCONCLUSIVE reason=budget_exceeded n_tested={n_tested}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_deaa2972.py", line 93, in <module>
    result = run_trial(seed)
              ~~~~~^~~~~~
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_deaa2972.py", line 50, in run_trial
    A = matrix_from_function(f, n)
        ~~~~~^~~~~~
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_deaa2972.py", line 25, in matrix_from_function
    A = [[f[i * (1 << n) + j] for i in range(1 << (n-1))] for j in range(1 << (n-1))]
          ~~~~~^~~~~~
IndexError: list index out of range

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test crashed before producing data, which means it did not complete its execution to provide results for the conjecture's support condition. | next: Re-run the test with a different seed or investigate the cause of the crash to ensure that the test can be completed successfully.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	14575
2	preregistration	ollama_remote	glm4:latest	0	0	9875
3	novelty	ollama_remote	glm4:latest	0	0	8490
4	novelty	ollama_remote	glm4:latest	0	0	8482
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	22936
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8297
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	16955

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11823
9	judge	ollama_remote	glm4:latest	0	0	16676

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 118109 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in research/audit/4cf209a9cba5.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/4cf209a9cba5.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/4cf209a9cba5.tar.gz (if generated)